

References

Research Papers Reviewed

1. Vision-based Fuzzy Control System for Smart Walkers

Authors: M. Chalak et al. Year: 2025 Contribution: Smart walker system using camera data and fuzzy logic for rehabilitation support.

2. Monitoring Walker Assistive Devices

Authors: V. Viegas et al. Year: 2018 Contribution: Load cell and optical sensor-based walker monitoring system.

3. Wearable Sensors and Rehabilitation Systems

Authors: S. Patel et al. Year: 2012 Contribution: Rehabilitation monitoring using wearable sensors.

4. Wearable Sensors and Machine Learning in Rehabilitation

Authors: S. Wei and Z. Wu Year: 2023 Contribution: Machine learning-based rehabilitation assessment.

5. Intelligent Diagnosis Using Shoe-Integrated Sensors

Authors: Z. Guo et al. Year: 2025 Contribution: Gait monitoring and rehabilitation prediction using machine learning.

6. Design and Implementation of Intelligent Crutches

Authors: R. K. Megalingam et al. Year: 2019 Contribution: Smart crutch with force sensing and rehabilitation monitoring.

7. Smart Crutches for Rehabilitation

Authors: Y. F. Chen et al. Year: 2018 Contribution: Instrumented crutches with vibration feedback and weight monitoring.

8. Assistive and Monitoring Multifunctional Smart Crutch

Authors: J. Yang et al. Year: 2019 Contribution: Smart crutch with monitoring and elderly assistance features.

9. Smart Crutches Using IoT

Authors: R. Varghese et al. Year: 2024 Contribution: GPS tracking and emergency assistance system.

10. Multi-Function Intelligent Crutch Implementation

Authors: Q. Zhang and S. Liu Year: 2022 Contribution: Obstacle detection, health monitoring, and mobile connectivity.

Total Papers Reviewed

- Research Papers Reviewed: 32
- Domain: Smart Crutch Rehabilitation Systems
- Technologies Studied:
 - Load Cells
 - HX711 Amplifier
 - ESP32
 - Bluetooth Communication
 - Mobile Applications
 - Machine Learning
 - IoT Monitoring
 - GPS Tracking
 - Rehabilitation Analytics

Research Gap Identified

Most existing smart crutch systems are expensive, complex, and lack an integrated low-cost rehabilitation monitoring solution with real-time feedback, mobile tracking, and machine learning-based recovery assessment.

Proposed Solution

Develop a low-cost Smart Rehabilitation Crutch using:

- ESP32
- Load Cell
- HX711
- Vibration Feedback
- Bluetooth Connectivity
- Mobile Application
- Machine Learning-Based Recovery Prediction